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Zhiqiang Zang

Education

2018–2024 **Ph.D.**, *The University of Texas at Austin (UT Austin)*, Austin, TX, USA
Software Engineering and Systems — Advisor: Milos Gligoric

2014–2018 **B.E.**, *Beijing University of Posts and Telecommunications (BUPT)*, Beijing, China
Telecommunication Engineering

Industry Experience

Feb 2024–Present **Research Scientist**, *Meta Platforms*

Oct 2026–Present Ads Training Infra, Vancouver, BC, Canada
Sparse optimization.

May 2026–Oct 2026 AI Trainers Platform

Dec 2025–May 2026 MSL (Meta Superintelligence Labs)
LLM profiling, performance observability and debugging.

Jun 2025–Dec 2025 AI Observability Platform
GPU performance debugging, LLM training profiling, AMD training support.

Feb 2024–Jun 2025 AI Infra Efficiency, Menlo Park, CA, USA
Peformance and efficiency tooling, Holistic Trace Analysis (HTA).

May–Aug 2023 **Software Engineer Intern**, *Uber Technologies*, San Francisco, CA, USA
Supported contracts in a Go static analyzer called NilAway (`https://github.com/uber-go/nilaway`) to reduce false positive errors.

May–Aug 2022 **Software Engineer Intern**, *Meta Platforms*, Bellevue, WA, USA
Migrated a large-scale data analytic engine onto a new environment to improve performance and maintainability.

May–Aug 2020 **Research Intern**, *Fujitsu Laboratories of America*, ~~Sunnyvale, CA, USA~~ Remote
Improved software program repair examples by minimizing buggy/fixed pairs of programs.

Jul–Aug 2019 **Software Engineer Intern**, *NIO*, Beijing, China
Developed and improved a physical and visual simulator for autonomous vehicles (Unreal Engine).

Patents

[1] Hiroaki Yoshida, **Zhiqiang Zang**, Mukul R. Prasad. Generation of Software Program Repair Examples. US11099817B1. Aug 24, 2021.

Publications

[7] **Zhiqiang Zang**, Fu-Yao Yu, Aditya Thimmaiah, August Shi, and Milos Gligoric. Java JIT Testing with Template Extraction. In *International Conference on the Foundations of Software Engineering*, pages 51:1–51:23, 2024.

[6] **Zhiqiang Zang**, Aditya Thimmaiah, and Milos Gligoric. JOG: Java JIT Peephole Optimizations and Tests from Patterns. In *International Conference on Software Engineering, Tool Demonstrations Track*, pages 11–15, 2024.

[5] **Zhiqiang Zang**, Aditya Thimmaiah, and Milos Gligoric. Pattern-Based Peephole Optimizations with Java JIT Tests. In *International Symposium on Software Testing and Analysis*, pages 64–75, 2023.

[4] **Zhiqiang Zang**, Fu-Yao Yu, Nathaniel Wiatrek, Milos Gligoric, and August Shi. JAttack: Java JIT Testing using Template Programs. In *International Conference on Software Engineering, Tool Demonstrations Track*, pages 6–10, 2023.

[3] **Zhiqiang Zang**, Nathaniel Wiatrek, Milos Gligoric, and August Shi. Compiler Testing using Template Java Programs. In *International Conference on Automated Software Engineering*, pages 23:1–23:13, 2022. **ACM SIGSOFT Distinguished Paper Award**.

[2] Pengyu Nie, Marinela Parovic, **Zhiqiang Zang**, Sarfraz Khurshid, Aleksandar Milicevic, and Milos Gligoric. Unifying Execution of Imperative Generators and Declarative Specifications. In *International Conference on Object-Oriented Programming, Systems, Languages, and Applications*, pages 217:1–217:26, 2020.

[1] Ben Buhse, Thomas Wei, **Zhiqiang Zang**, Aleksandar Milicevic, and Milos Gligoric. VeDebug: Regression Debugging Tool for Java. In *International Conference on Software Engineering, Tool Demonstrations Track*, pages 15–18, 2019.

Selected Projects

JAttack `https://github.com/EngineeringSoftware/jattack`
A template-based compiler testing framework that 1) accepts as input a template Java program written in a domain-specific language in Java; 2) executes the template to generate concrete programs and 3) tests Java JIT compilers using the generated programs. JavaParser, OW2 ASM, DSL, JIT, Java. Team Project. Discovered CVEs: CVE-2020-14792, CVE-2022-21305, CVE-2023-22044, CVE-2023-22045.

JOG `https://github.com/EngineeringSoftware/jog`
A framework that facilitates developing Java JIT peephole optimizations: 1) accepts as input a pattern written in the same way that tests are written; 2) translates the pattern into C/C++ code as a JIT optimization pass, and generates Java tests for the optimization; 3) detects shadow relation between a pair of optimizations. JavaParser, Z3, DSL, JIT, Java. Team Project. 7 pull requests have been integrated into the master branch of OpenJDK.

VeDebug `https://github.com/EngineeringSoftware/VeDebug`
A Java debugging tool that 1) automatically sets breakpoints where the current execution diverges from the previously captured one, and 2) provides video player features e.g., speed up/slow down the replay. Bytecode instrumentation(OW2 ASM, javaagent), Java. Team project.

Technical Skills

- Programming Languages: Java, Bash, Python (everyday), C, Go, C++
- Tools: Emacs (everyday), Git/Mercurial (everyday), OW2 ASM, JavaParser, Holistic Trace Analysis (HTA), PyTorch